

Deep Learning-Based Audio Language Model (ALM) for Multimodal Speech and Non-Speech Understanding

Adithya K.A

Department of CIT

Presidency University,

Yelahanka,Bangalore-560064, India

koodathadithya@gmail.com

K Sai Lakshmi

Department of CIT

Presidency University,

Yelahanka,Bangalore-560064, India

lakshminaveen0927@gmail.com

Anand M.M

Department of CIT

Presidency University,

Yelahanka,Bangalore-560064, India

anandmm0628@gmail.com

Kaavya Gaur

Assistant Professor

School of Computer Science and Engineering,

Presidency University,

Yelahanka,Bangalore-560064, India

kaavyagaur@presidencyuniversity.in

Abstract—This paper presents MINDSTREAM, a multimodal Audio Language Model (ALM) designed to integrate speech, environmental audio, visual input, and location data for real-time contextual understanding. Unlike conventional systems that treat non-speech audio as background noise, the proposed system utilizes both speech and non-speech signals as meaningful contextual inputs. The architecture adopts a modular design with deep learning-based encoders and a multithreaded processing framework to enable efficient multimodal fusion and low-latency interaction. The system incorporates a multimodal reasoning engine supported by external AI services such as Gemini to generate context-aware responses.

The proposed system is evaluated using scenario-based real-time testing across various application conditions, focusing on functional performance, responsiveness, and multimodal integration. The observations indicate that combining multiple modalities enhances contextual interpretation compared to single-modality processing. The inclusion of location-aware capabilities further improves system applicability in real-world scenarios such as navigation and emergency assistance..

Index Terms—Audio Language Model, Multimodal AI, Speech and Sound Understanding, Image Recognition, Location Awareness, Real-Time Processing, Deep Learning.

I. INTRODUCTION

Conversational multimodal intelligence has emerged as an important area of research in artificial intelligence, focusing on the integration and processing of multiple forms of input such as speech, environmental sounds, visual data, and spatial context. With the growing demand for intelligent assistants, there is an increasing need for systems capable of multimodal

perception and reasoning, enabling them to effectively utilize diverse input modalities.

Most existing state-of-the-art systems are limited to single-modality processing, such as speech recognition or text-based interaction. Even in systems that process audio, environmental sounds are often treated as background noise rather than meaningful contextual information. As a result, such systems struggle to operate effectively in complex, dynamic environments that require simultaneous interpretation of audio, visual, and spatial data. This limitation reduces their ability to provide contextually relevant and accurate responses.

To address these challenges, this paper proposes MINDSTREAM (Multimodal Intelligent Neural Device for Streaming, Thinking, and Real-Time Environment Awareness), a multimodal Audio Language Model (ALM)-based system. The proposed architecture integrates speech recognition, environmental sound analysis, computer vision-based scene understanding, and location-aware processing into a unified framework. By combining these modalities, the system enhances situational awareness and improves response relevance.

The system is designed using a modular architecture supported by multithreaded processing, enabling efficient handling of heterogeneous data streams in real-time. Unlike conventional approaches, the proposed system treats environmental audio as a valuable source of contextual information, leading to more meaningful and accurate responses. The system is evaluated through scenario-based testing across various applications, including assistive technologies, intelligent environments, surveillance, and emergency response systems.

Contributions:

The main contributions of this work are summarized as follows:

- A unified multimodal framework integrating speech, environmental audio, visual input, and location data for comprehensive contextual understanding.
- A multithreaded processing architecture enabling low-latency, real-time interaction.
- A cross-modal reasoning approach for generating context-aware responses from heterogeneous inputs.
- A scenario-based system evaluation demonstrating the effectiveness of multimodal integration in practical real-world applications.

II. LITERATURE SURVEY

Recent advancements in multimodal artificial intelligence have enabled systems to process and integrate diverse data sources such as audio, visual, and contextual information for enhanced human-computer interaction. The emergence of large-scale multimodal models, such as Gemini [10], has significantly accelerated research in unified reasoning across heterogeneous data modalities. In the domain of speech and audio processing, deep learning-based approaches have achieved substantial improvements in robustness and accuracy. Large-scale models such as Whisper [4] demonstrate the effectiveness of weakly supervised learning for speech recognition across diverse environments. Furthermore, audio-visual speech recognition techniques [5] improve performance by combining acoustic and visual cues, particularly in noisy conditions. Recent developments such as AudioPaLM [6] extend these capabilities by enabling unified modeling of speech and environmental audio. Multimodal learning frameworks provide structured approaches for integrating heterogeneous data sources. Early work by Liang et al. [1] introduced multistage fusion techniques for multimodal language analysis. Subsequent research [2], [3] further improved cross-modal representation learning and fusion strategies, enabling better contextual understanding. More recent contributions [13], [14] focus on advanced multimodal alignment and classification, improving performance in complex real-world scenarios. In computer vision, transformer-based architectures such as Vision Transformers [7] have significantly improved image recognition and scene understanding. Vision-language models [8] extend these capabilities by enabling semantic interpretation of visual data through natural language. Systems such as LLaVA [9] demonstrate the ability to perform interactive reasoning across visual and textual inputs. Recent developments in multimodal AI systems [11], [12] highlight the importance of integrating multiple data streams for real-world applications such as smart environments and intelligent assistants. These studies show that combining different modalities leads to improved contextual awareness and decision-making. Location-based systems play a crucial role in enabling context-aware services. Techniques such as Wi-Fi-based indoor localization [15], fingerprint-based positioning [16], and visible light positioning systems [17] provide accurate

spatial awareness in both indoor and outdoor environments. These approaches emphasize the importance of incorporating location data into multimodal systems for enhanced real-world interaction. Despite these advancements, many existing systems focus on limited modality combinations and lack unified real-time reasoning capabilities. There remains a need for systems that integrate speech, environmental audio, visual input, and spatial context into a single framework. The proposed MINDSTREAM architecture addresses this gap by providing a comprehensive multimodal integration and real-time reasoning system.

TABLE I
SUMMARY OF LITERATURE REVIEW

Area	Contribution	Ref.
Speech & Audio	Improves speech and audio understanding using large models	[4], [6]
Audio-Visual	Combines audio and visual cues for noisy environments	[5]
Multimodal	Fusion of multiple data types for better understanding	[1]–[3], [13], [14]
Vision	Transformer-based image understanding	[7]
Vision-Language	Image understanding using natural language	[8], [9]
Multimodal AI	Unified reasoning across modalities	[10]–[12]
Location	Context-aware services using spatial data	[15]–[17]

III. PROPOSED MULTIMODAL SYSTEM ARCHITECTURE AND WORKFLOW

The proposed MINDSTREAM system is designed as a real-time multimodal framework that integrates speech, environmental audio, visual input, and location-based contextual information into a unified reasoning pipeline. Unlike traditional systems that process a single modality or treat non-speech audio as noise, the proposed approach utilizes all available inputs as meaningful contextual signals. The architecture follows a modular and scalable design, enabling efficient processing of heterogeneous inputs while maintaining low latency.

A. System Overview

The overall system is organized into four primary modules:

- Audio Processing Module – Handles speech and environmental sounds
- Visual Processing Module – Processes image data and extracts scene information
- Contextual Module – Incorporates location and situational awareness
- Multimodal Reasoning Engine – Performs unified contextual reasoning

These modules operate concurrently and are connected through a multimodal fusion layer, ensuring that each modality contributes to the final decision-making process.

B. Input Acquisition and Preprocessing

The system captures multiple input streams simultaneously:

- **Speech Input:** User voice commands are captured through a microphone and converted into text using speech recognition models.
- **Environmental Audio:** Non-speech sounds such as alarms and traffic noise are analyzed using deep learning-based acoustic models.
- **Visual Input:** Image frames from the camera are processed using computer vision techniques such as object detection and scene recognition.
- **Location Data:** GPS or API-based services provide spatial context for better interpretation of user queries.

Each modality is preprocessed independently to extract relevant features before integration.

C. Multimodal Fusion Mechanism

The outputs from all preprocessing modules are combined in a multimodal fusion layer to form a unified representation. This process involves feature-level and semantic-level integration. Temporal synchronization is maintained using timestamp alignment to ensure consistency across inputs.

By combining multiple modalities, the system reduces ambiguity and enhances contextual understanding, especially in dynamic environments.

D. Multimodal Reasoning Engine

The fused representation is processed by a multimodal reasoning engine powered by advanced AI models.

The reasoning engine performs:

- Interpretation of multimodal inputs
- Maintenance of conversational context
- Generation of context-aware responses

This enables more accurate and intelligent decision-making compared to traditional unimodal systems.

E. Workflow Execution

The system workflow supports real-time interaction with minimal latency:

- Simultaneous acquisition of audio, visual, and location data
- Parallel preprocessing using multithreading
- Fusion into a unified multimodal representation
- Contextual reasoning and response generation
- Output delivery in text and optional speech format

F. System Capabilities and Advantages

The proposed system offers:

- Real-time performance through parallel processing
- Enhanced contextual awareness using multimodal fusion
- Improved response accuracy
- Scalability and modular design

The architecture can be extended for applications such as assistive technologies, smart environments, surveillance systems, and intelligent user interfaces.

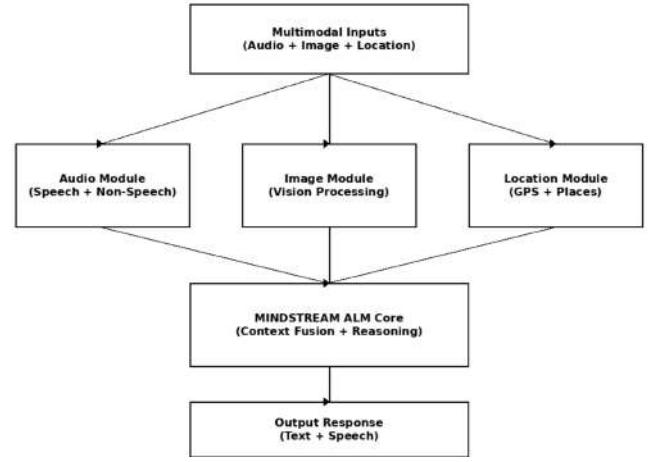


Fig. 1. Workflow Execution

IV. IMPLEMENTATION

The proposed Multimodal Audio Language Model (ALM) solution requires a number of supporting technologies and software libraries to be incorporated into the solution to process speech, ambient sounds, images and location information in real-time. The system has been implemented in Python because it is versatile, scalable, and boasts of enormous availability of software libraries in the area of artificial intelligence (AI), machine learning and multimodal processing. The language is very strong in enforcement of speech recognition, sound analysis in the environment, algorithm of computer vision as well as integrations of APIs. The Python language is developed in a modular way hence is scalable in future functions of AI and upgrades.

A. Image Processing and Visual Analysis

The system incorporates built-in camera features on gadgets like laptops or smartphones to take real-time pictures. Gemini 2.5 Pro can be employed as a sophisticated model of multimodal processing because it allows data to be easily processed concerning image preprocessing like pre-processing activity resizing, converting the image format, normalization, and even enhancing it in one pipeline. Moreover, it provides advanced computer vision features like object detection, image segmentation, caption generation, and optical character recognition and, thus, makes it easier to analyze images comprehensively. This allows freeflow of work where the images are effectively handled to make the best use of them with a well-integrated system thus without the need to have further preprocessing libraries.

B. Speech Interaction

Voice input can be supported with the help of the advanced speech-to-text communication system allowing to support the audio inputs without any additional encoding systems like base64 encoding. The audio data are recorded in real-time and subsequently processed to result in a transcription that is both accurate when the environment is not favorable or should

the usage be hands-free. Text information is then channeled through the ALM reasoning engine so as to generate context based information based on the input information and the context of the environment. The system is in a position to deliver smooth voice interaction and this ease of handling the audio information.

C. Multithreading and Real-Time Processing

In order to have the required responsiveness of the system and real-time performance, they apply a concept of multithreading as a way of dealing with multiple streams of data simultaneously. During concurrent processing, audio data, speech and non-speech, visual data and location data are processed. Every functional domain is deployed as a thread. Examples of these threads include audio capture and processing, image capture and processing, location information retrieval, and ALM-based reasoning. In this way, high performance in real-time is performed efficiently.

D. Development Environment

The given system is developed on the Python language. The language possesses a wide range of libraries of audio processing, computer vision, and API integrations. The proposed system consists of advanced AI services, such as multimodal models and APIs, to ensure that it is competent in speech recognition, analyzing environment sound, computer vision, etc. The proposed architecture is versatile and can be executed on various platforms on desktop or mobile environment SDKs and APIs. Regular package managers are being used to manage dependency. Therefore, the suggested system can be extended and incorporated into future developments in AI technologies.

V. SYSTEM EVALUATION

The MINDSTREAMs system was evaluated in various modalities, including speech recognition, environmental sounds, images and multimodal inference. To validate the degree of efficiency of the system in real-time use, quantitative measures of measurement were applied.

A. Evaluation Approach

The proposed system is evaluated using a scenario-based real-time testing approach rather than a traditional dataset-based evaluation. Since the system relies on external AI services such as Google Gemini, the focus of evaluation is on practical system performance, responsiveness, and multimodal integration in real-world conditions.

B. Test Scenarios

The system was tested across multiple real-world scenarios, including:

- Voice command recognition in quiet and moderately noisy environments
- Detection of environmental sounds such as alarms and traffic
- Image-based scene understanding using camera input
- Location-based queries such as identifying the nearest police station or hospital

These scenarios were selected to reflect practical use cases in everyday interaction and emergency situations.

C. Observations

The system demonstrated reliable performance across different modalities. Speech inputs were accurately recognized under normal conditions, while environmental sounds were effectively identified and incorporated into contextual understanding. The visual module performed well in detecting objects and interpreting scenes in well-lit environments. The integration of location data significantly enhanced system capability. For example, when a user queried “nearest police station,” the system utilized spatial data along with external APIs to retrieve relevant nearby services and generate an appropriate response.



Fig. 2. proposed multimodal framework

D. Performance Analysis

The use of a multithreaded architecture enabled simultaneous processing of audio, visual, and contextual inputs, resulting in smooth real-time interaction. The system exhibited minimal perceptible delay; however, response time is influenced by external factors such as API processing time and network latency.

E. Comparison with Baseline Systems

To evaluate the effectiveness of the proposed MINDSTREAM system, a qualitative comparison was performed against existing systems that operate on limited modalities. Traditional pipelines such as speech-only systems (e.g., Whisper-based models) and vision-language models (e.g., LLaVA) were considered as baselines.

Unlike these systems, which process individual or dual modalities, the proposed system integrates speech, environmental audio, visual input, and location context in a unified framework. This enables improved situational awareness and more context-aware responses in real-time scenarios.

TABLE II
COMPARISON WITH BASELINE SYSTEMS

System	Modalities	Real-Time	Context Awareness
Whisper-based System	Audio (Speech)	Yes	Low
LLaVA	Vision + Text	Partial	Medium
Proposed MINDSTREAM	Audio + Vision + Location	Yes	High

F. Limitations

Certain limitations were observed during testing. Performance may degrade in high-noise environments affecting speech recognition accuracy. Similarly, low-light conditions can reduce the effectiveness of visual processing. Additionally, dependence on external APIs introduces reliance on network connectivity and service availability.

G. Overall Evaluation

Overall, the system demonstrates the effectiveness of multimodal integration for real-time contextual understanding. By combining speech, environmental audio, visual input, and location data, the system provides more relevant and context-aware responses compared to single-modality approaches. The inclusion of location-aware functionality further enhances its applicability in real-world scenarios such as navigation and emergency assistance.

TABLE III
SYSTEM FEATURES AND PERFORMANCE

Feature	Description	Performance
Device Coverage	Laptop, smartphone camera	Reliable capture
Location Tracking	GPS integration	Accurate spatial context
Image Use Cases	Object detection, scene	Strong performance
Image Preprocessing	Multimodal preprocessing	Better accuracy
Visual Analysis	Detection, segmentation	Slight drop in low-light
Multimodal Interaction	Audio + image + location	Better awareness
Response Generation	Context-aware replies	Timely outputs
Audio Processing	Speech + sound classification	Noise sensitive
Performance	Multithreading	Low delay
User Feedback	Accessibility use	High engagement
Limitations	Low-light, noise issues	Needs improvement

VI. ETHICAL CONSIDERATIONS

The system processes sensitive data such as audio, visual input, and location information. To ensure user privacy, data collection is performed only with user consent, and no personally identifiable information is stored. The system is designed to promote responsible and ethical use of AI technologies.

VII. CONCLUSION

The proposed research took into consideration an architecture of designing a prototype of a Multimodal ALM in the real-time interaction by incorporating speech, environmental sounds, vision and location context, namely MINDSTREAM (Multimodal Intelligent Neural Device for Streaming, Thinking & Real-time Environment Awareness). It should be mentioned that the level of intelligence in this system is based on highly developed multimodal models capable to conduct contextual reasoning with the help of different data sources and interpret the intentions and the environment of the user properly. Thus, the model will be capable of analyzing data

regarding acoustic scenes and at the same time processing images and their properties. The model comprises of object recognition, object detection, segmentation, captioning and optical character recognition. Moreover, it is capable of executing a task of decoding verbal commands or ambient noise. As discussed in the earlier part, the model offers multithreaded processing that ensures that the latency is low and enables one to work with multiple input streams simultaneously. Therefore, the experiments conducted during the assessment of the efficiency of the suggested system demonstrated the usefulness of the system in other practical implementations. Despite this achievement, there are a few drawbacks of the system in extreme conditions of low light environment, high noises, and higher levels of computational complexity in high resolution inputs. The planned improvements would be the upgrading of the existing solution with the video data to understand the temporal context of the picture, interoperability with cross-platforms, and contextual memory to enhance the interaction with the user of the long-term interaction. Additionally, the use of the state-of-the-art tools like transformers as well as the corresponding domain data is likely to present an added impetus in the system. In addition, the suggested system, MINDSTREAM, demonstrates the huge capabilities of multimodal fusion technology regarding obtaining a superior quality situational awareness through speech data, sound cues, visual data, and spatial situation. By applying such a technique, much more powerful intelligent assistants can be developed which will allow natural and high-quality interaction with end users in real-time environment.

REFERENCES

- [1] P. P. Liang *et al.*, “Multimodal language analysis with recurrent multistage fusion,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2018.
- [2] S. Mai, H. Hu, and S. Xing, “Modality to modality translation: An adversarial representation learning and graph fusion network for multimodal fusion,” *IEEE Transactions on Multimedia*, vol. 23, pp. 109–122, 2021.
- [3] T. Mittal, A. Bera, and D. Manocha, “Multimodal and context-aware emotion perception model with multiplicative fusion,” *IEEE Multimedia*, vol. 28, no. 2, pp. 67–75, 2021.
- [4] A. Radford *et al.*, “Robust speech recognition via large-scale weak supervision,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 2022.
- [5] Y. Afouras, J. S. Chung, and A. Zisserman, “Deep audio-visual speech recognition,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 12, pp. 8717–8732, 2022.
- [6] P. Liang *et al.*, “AudioPaLM: A large language model for speech and audio,” *Proc. IEEE ICASSP*, 2023.
- [7] A. Dosovitskiy *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *Proc. IEEE/CVF CVPR*, 2021.
- [8] H. Xu *et al.*, “Vision-language pre-training for multimodal understanding: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [9] H. Liu *et al.*, “Visual instruction tuning,” *Proc. IEEE/CVF CVPR*, 2023.
- [10] Google Research, “Gemini: A family of highly capable multimodal models,” 2023.
- [11] L. Liao *et al.*, “Multimodal AI for big data analytics,” *IEEE Big Data Workshop (MMAI)*, 2023.
- [12] “Advancing multimodal AI research and applications,” *IEEE ICDM Workshops (ICDMW)*, 2025.
- [13] Y. Ma *et al.*, “A 3D self-awareness diffusion network for multimodal classification,” *IEEE Transactions on Multimedia*, vol. 27, pp. 3462–3475, 2025.

- [14] M. Matar *et al.*, “Optimizing multimodal feature alignment for enhanced text-to-video retrieval,” *Proc. IEEE ICIT*, 2025, pp. 242–246.
- [15] J. Wang, Y. Chen, and L. Zhang, “Deep learning-based indoor localization using Wi-Fi signals,” *IEEE Access*, vol. 8, pp. 121–132, 2020.
- [16] S. He and S.-H. G. Chan, “Wi-Fi fingerprint-based indoor positioning: Recent advances and comparisons,” *IEEE Communications Surveys & Tutorials*, vol. 18, no. 1, pp. 466–490, 2020.
- [17] Y. Zhuang *et al.*, “A survey of positioning systems using visible LED lights,” *IEEE Communications Surveys & Tutorials*, vol. 20, no. 3, pp. 1963–1988, 2021.